E-R MODEL

EXAMPLES & PRACTICE QUESTIONS

How to Draw an ER Diagram ?

- **Identify Entities:** The very first step is to identify all the Entities. Represent these entities in a Rectangle and label them accordingly.
- **Identify Relationships**: The next step is to identify the relationship between them and represent them accordingly using the Diamond shape. Ensure that relationships are not directly connected to each other.
- **Add Attributes**: Attach attributes to the entities by using ovals. Each entity can have multiple attributes (such as name, age, etc.), which are connected to the respective entity.

How to Draw an ER Diagram ?

- **Define Primary Keys:** Assign primary keys to each entity. These are unique identifiers that help distinguish each instance of the entity. Represent them with underlined attributes.
- **Remove Redundancies**: Review the diagram and eliminate unnecessary or repetitive entities and relationships.
- **Review for Clarity**: Review the diagram make sure it is clear and effectively conveys the relationships between the entities.

Example1 : University Management System

A university wants to maintain records of students, courses, instructors, and departments. Students enroll in courses, instructors teach courses, and departments offer courses. Each student belongs to one department. *Draw the E–R diagram for the above system.*

BRIEF INFORMATION

- A University contains many departments
- Each department can offer any number of courses
- Many instructors can work in a department
- An instructor can work only in one department
- For each department there is a Head
- An instructor can be head of only one department
- Each instructor can take any number of courses
- A course can be taken by only one instructor
- A student can enroll for any number of courses
- Each course can have any number of students

Step 1 : Identify the Entities

- From the statements given, the entities are

1. Department
2. Course
3. Instructor
4. Student

Step2: Identify the relationship

1. One department offers many courses. But one particular course can be offered by only one department. hence the cardinality between department and course is One to Many (1:N)
2. One department has multiple instructors . But instructor belongs to only one department. Hence the cardinality between department and instructor is One to Many (1:N)
3. One department has only one head and one head can be the head of only one department. Hence the cardinality is one to one. (1:1)
4. One course can be enrolled by many students and one student can enroll for many courses. Hence the cardinality between course and student is Many to Many (M:N)

5. One course is taught by only one instructor. But one instructor teaches many courses. Hence the cardinality between course and instructor is Many to One (N :1)

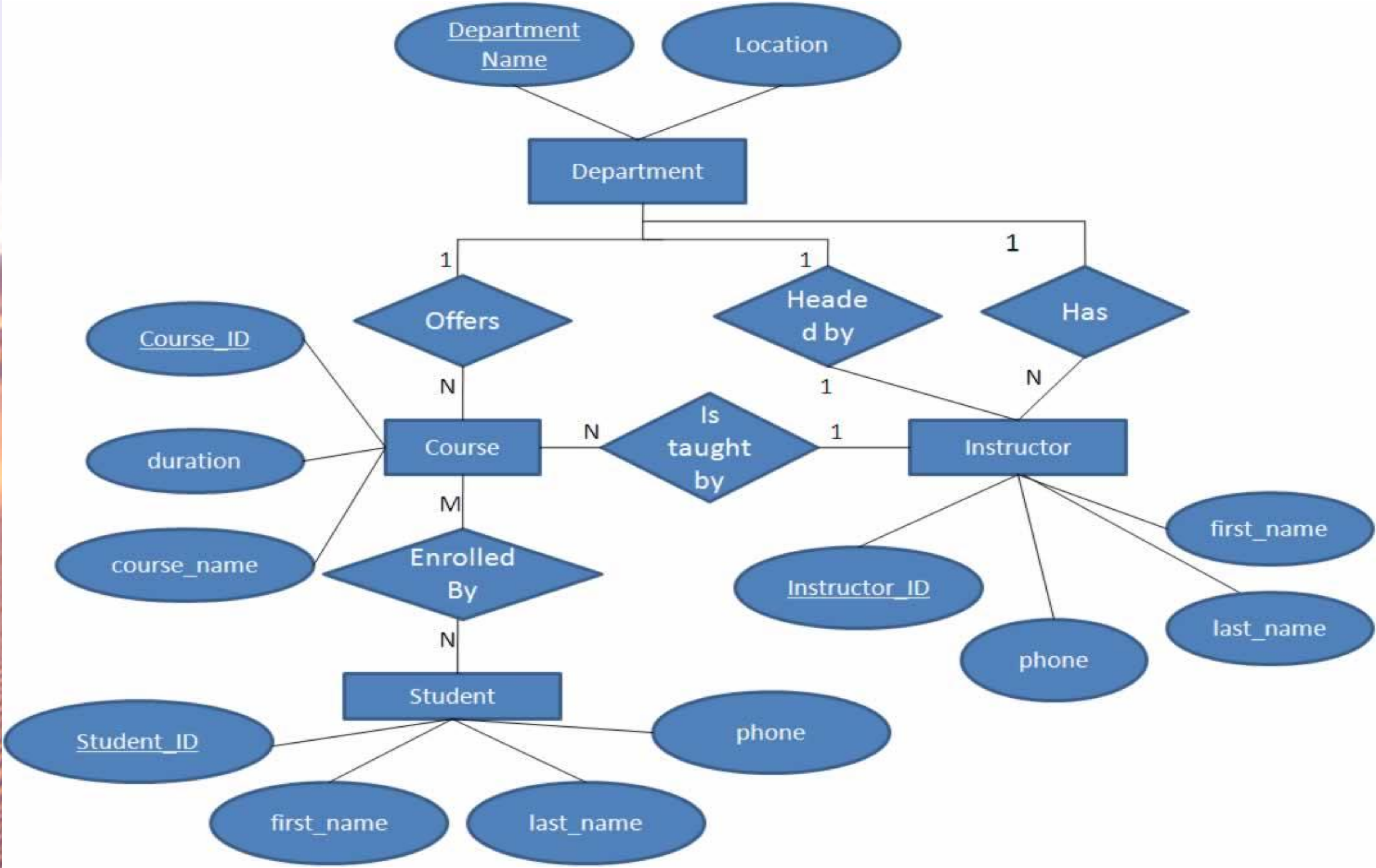
Step 3: Identify the key attributes

- "Departmen_Name" can identify a department uniquely. Hence Department_Name is the key attribute for the Entity "Department".
- Course_ID is the key attribute for "Course" Entity.
- Student_ID is the key attribute for "Student" Entity.
- Instructor_ID is the key attribute for "Instructor" Entity.

Step 4: Identify other relevant attributes

- For the department entity, other attributes are location
- For course entity, other attributes are course_name,duration
- For instructor entity, other attributes are first_name, last_name, phone
- For student entity, first_name, last_name, phone

Step 5: Draw complete ER diagram



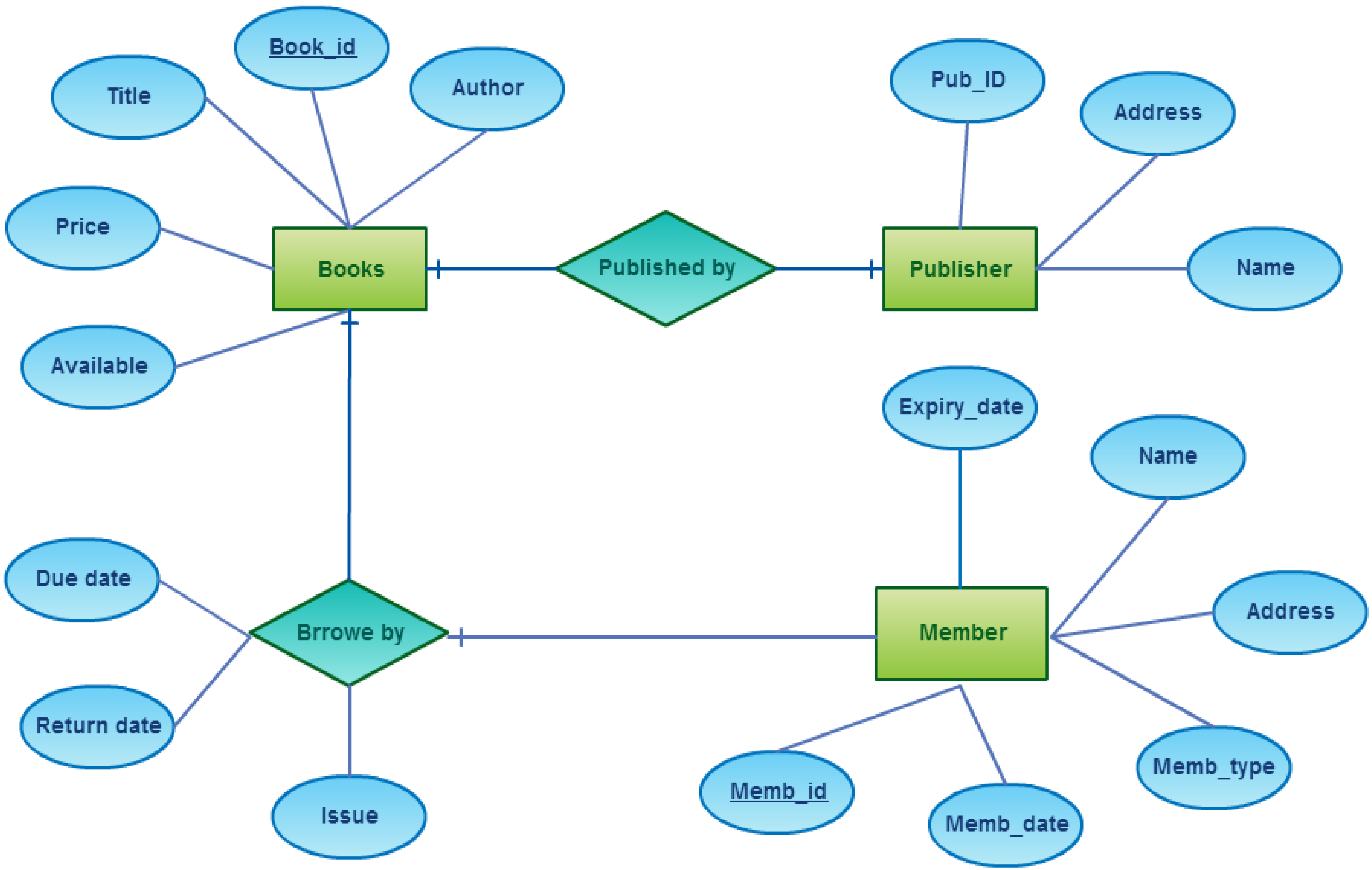
EXAMPLE 2:E-R DIAGRAM

Library Management System

A library wants to design a database to manage its daily operations. The library maintains information about books, publishers, and members. The library keeps a record of book issues and returns.

Design an E–R diagram for the Library Database Management System.

E-R Diagram of Library Management System



PRACTICE QUESTIONS

1. Railway Reservation System

A railway reservation system stores information about passengers, trains, tickets, and seats.
Passengers book tickets for trains and seats are allocated accordingly. *Draw the E–R diagram for the railway reservation system.*

2. Company Project System

A company maintains records of employees, departments, and projects. Employees work on projects and are assigned to departments. *Draw the E–R diagram for the company project system.*

PRACTICE QUESTIONS

3. Hospital Information System

A hospital maintains information about doctors, patients, departments, and treatments. Doctors treat patients, and treatments are recorded with relevant details. *Draw the E–R diagram for the hospital system.*

4. Online Shopping System

An online shopping platform maintains data about customers, orders, products, and payments.

Customers place orders, orders contain products, and payments are made for orders. D*raw the E–R diagram for the above system.*

PRACTICE QUESTIONS

5. Hotel Booking System

A hotel booking system maintains details of hotels, rooms, guests, and bookings.

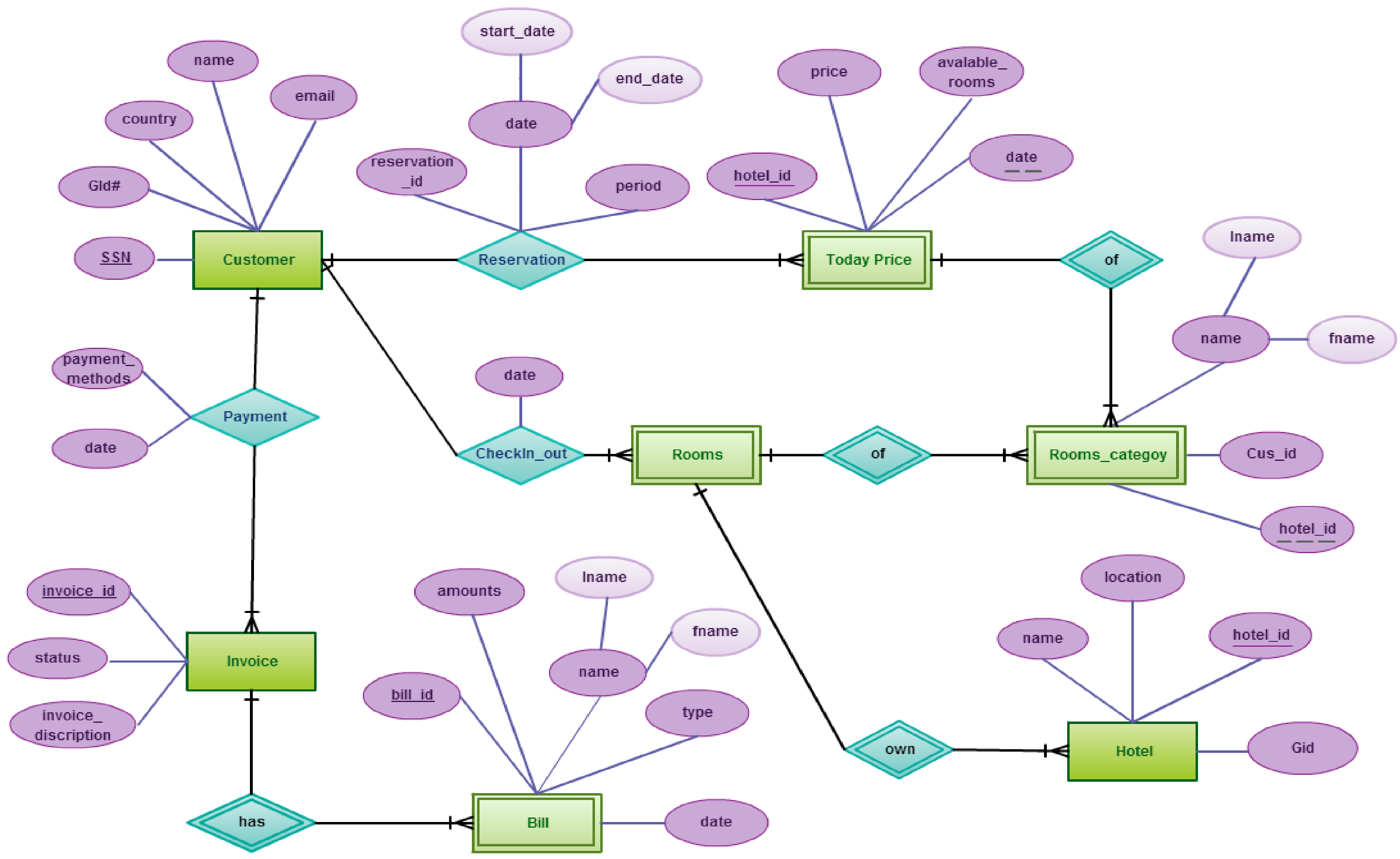
Guests book rooms and payment details are recorded. D*raw the E–R diagram for the hotel booking system.*

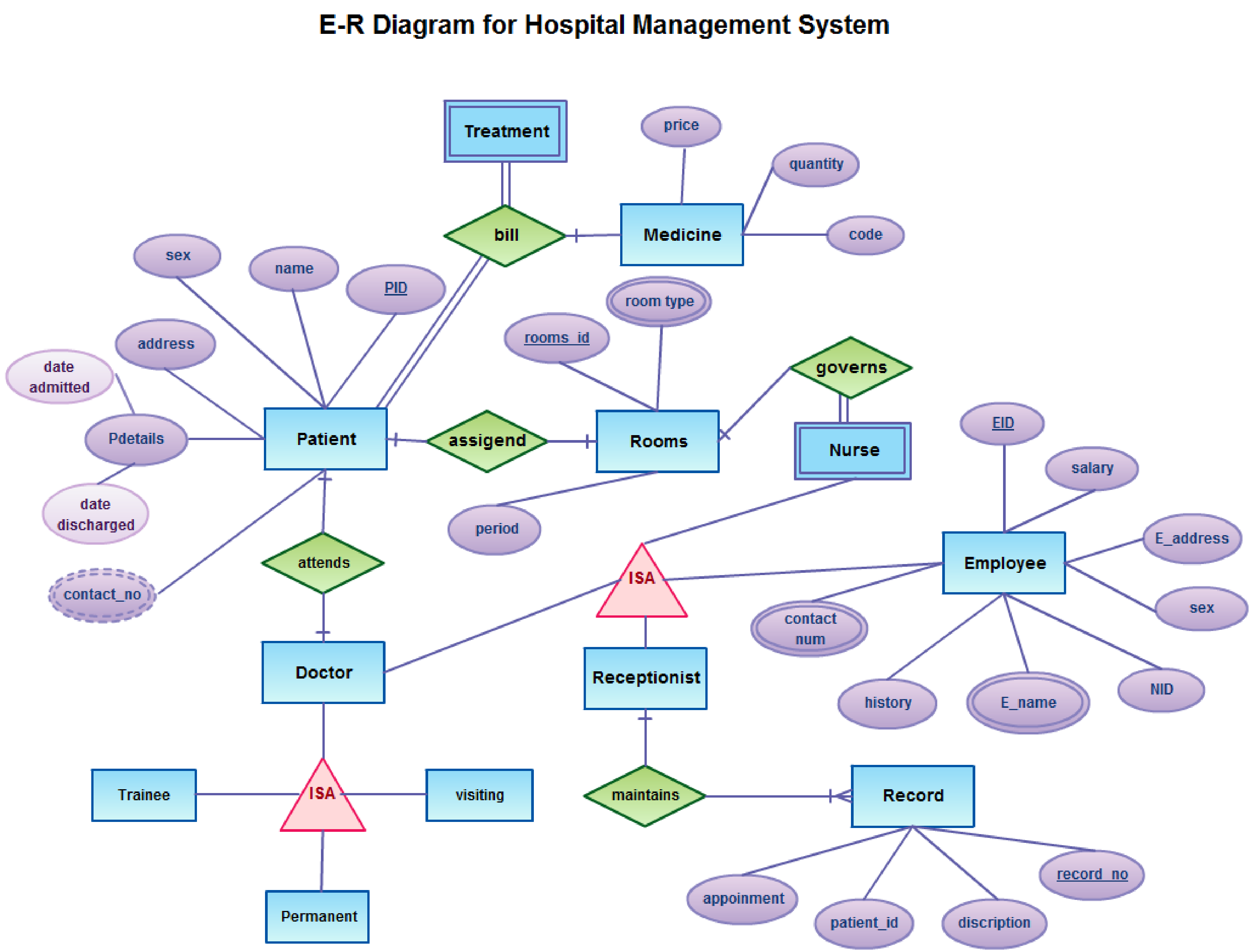
6. College Examination System

A college manages data related to students, subjects, examinations, and results. Students appear in examinations and results are declared for each subject.

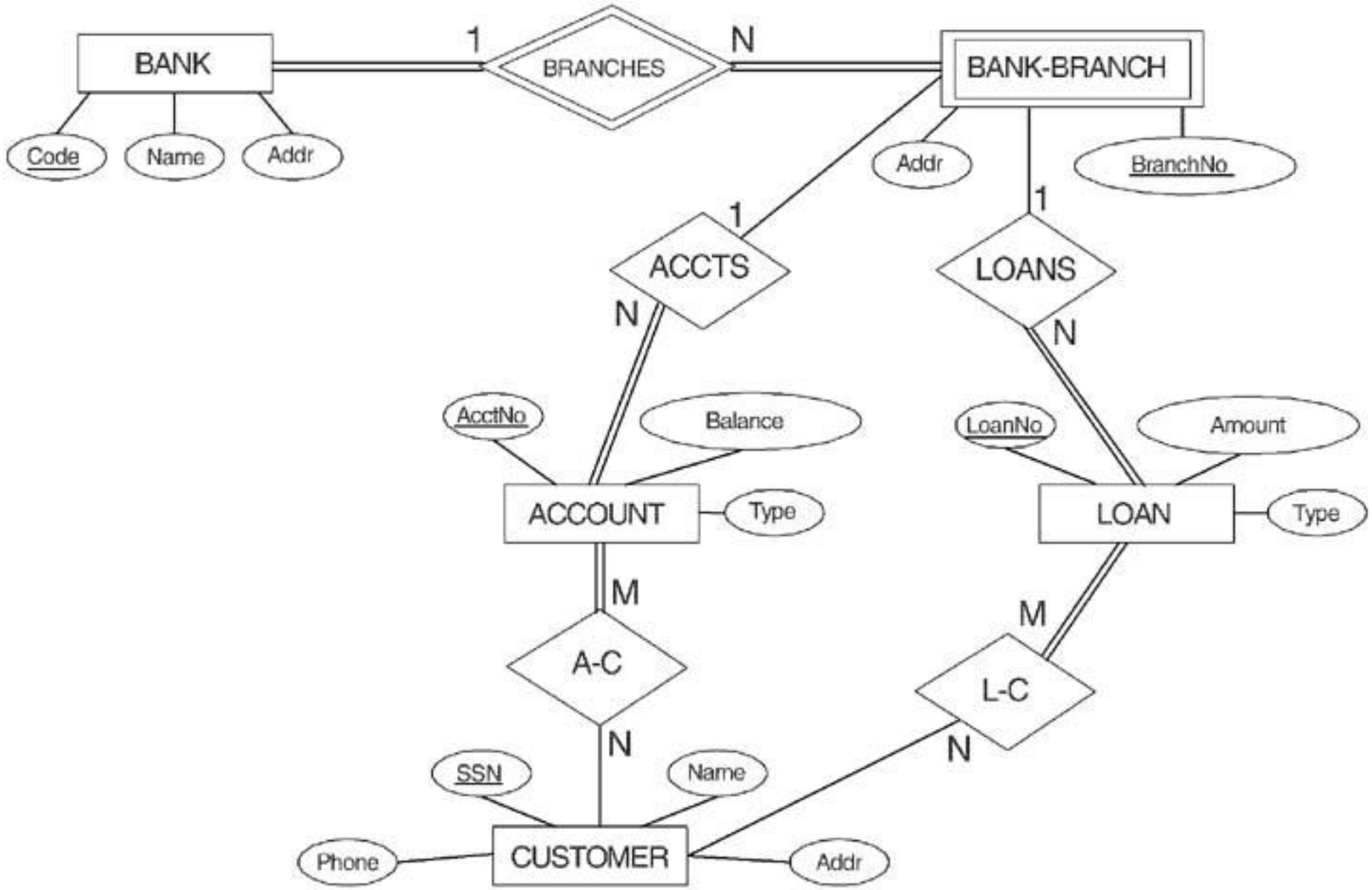
Draw the E–R diagram for the college examination system.

E-R Diagram for Hotel Management System




E-R Diagram for Hospital Management System
Treatment
price
quantity
bill
Medicine
code
sex
name
PID
room type
rooms_id
address
governs
date admitted
Patient
assigend
Rooms
Nurse
EID
Pdetails
salary
date discharged
period
attends
ISA
Employee
E_address
contact_no
contact num
sex
Doctor
Receptionist
history
E_name
NID
Trainee
ISA
visiting
maintains
Record
record_no
appoinment
patient_id
discription
Permanent

ER DIAGRAM FOR A BANK DATABASE



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